Chapter 8 solutions

CH08-E1

1. T_1 linear:

$$T_1(x, y) = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Every component is a linear form.

2. Since $T_2(0, 0) = (0, 1) \neq (0, 0)$, the map is nonlinear.

3. T_3 linear:

$$T_3(\alpha p + \beta q)(t) = t(\alpha p + \beta q)'(t) = \alpha t p'(t) + \beta t q'(t).$$

Also $tp'(t)$ still belongs to $\mathcal{P}_2$, so the map has the stated codomain.

4. Squaring is nonlinear. Set $f = g = 1$.

$$T_4(f + g) = 4 \neq 1 + 1 = T_4(f) + T_4(g).$$

CH08-E2

1.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

2. Solve

$$x + 2y - z = 0, \qquad x - y + 2z = 0.$$

Subtraction gives $3y - 3z = 0$, hence $y = z$ and $x = -z$. Therefore

$$N(T) = \operatorname{span}\{(-1, 1, 1)^T\}, \qquad \text{nullity} = 1.$$

3. The columns $a_1 = (1, 1)^T, a_2 = (2, -1)^T$ are independent, giving rank 2 and

$$C(T) = \mathbb{R}^2.$$

These two columns form an image basis.

4.

$$1 + 2 = 3 = \dim \mathbb{R}^3,$$

This verifies rank-nullity.

5. A nonzero kernel rules out injectivity. The image fills the codomain, proving surjectivity.

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CH08-E3

1. Input basis:

$$v_1 = (1,1)^T, \qquad v_2 = (1,-1)^T.$$
$$T(v_1) = (3,1,1)^T, \qquad T(v_2) = (-1,-1,1)^T.$$

Standard output coordinates:

$$M_1 = \begin{bmatrix} 3 & -1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

2. New output basis:

$$B_{\text{out}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad B_{\text{out}}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$
$$M_2 = B_{\text{out}}^{-1} M_1 = \begin{bmatrix} 2 & 0 \\ 0 & -2 \\ 1 & 1 \end{bmatrix}.$$

3. Here B_{in} and B_{out} denote the column matrices of the input basis and the output basis in part 2:

$$M_2 = B_{\text{out}}^{-1} A B_{\text{in}}.$$

The first column means

$$T(v_1) = 2w_1 + w_3.$$

The second means

$$T(v_2) = -2w_2 + w_3.$$

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CH08-E4

1. Trace 5 and determinant 6 give eigenvalues 3 and 2 in this two-dimensional case.

$$\lambda = 3: \quad (A - 3I)x = \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} x = 0,$$

Choose $v_1 = (2, 1)^T$.

$$\lambda = 2: \quad (A - 2I)x = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} x = 0,$$

Choose $v_2 = (1, 1)^T$.

2.

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

3.

$$B^{-1}AB = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}.$$

Trace 5 and determinant 6 agree with the original matrix.

4. Scaling each v_i by $s_i \neq 0$ gives

$$B' = B\,\mathrm{diag}(s_1, s_2).$$

The diagonal scaling commutes with the eigenvalue matrix and cancels with its inverse, leaving

$$B'^{-1}AB' = \mathrm{diag}(3, 2).$$

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CH08-E5

1. Input $(1, t, t^2)$, output $(1, t)$:

$$M_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

2. The kernel is the constant polynomials and the image is $\mathcal{P}_1$.

$$N(D) = \text{span}\{1\}, \qquad C(D) = \text{span}\{1, t\}.$$

Thus $3 = 1 + 2$, as rank-nullity requires.

3. Set $s = t - 1$.

$$D(1) = 0, \qquad D(s) = 1, \qquad D(s^2) = 2s = -2 + 2t.$$

Hence

$$M_2 = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}.$$

4. The matrices are 2×3, so similarity is not defined for them. They represent the same map in different domain bases, related by a two-basis conversion.

CH08-E6

1. The triangular matrix has eigenvalue 0 with algebraic multiplicity 2.

$$Nx = 0$$

The equation imposes $x_2 = 0$, so the eigenspace is

$$\text{span}\{(1, 0)^T\}.$$

2. Generalized vector:

$$w = (0, 1)^T, \qquad Nw = (1, 0)^T.$$

3. The given matrix is already a Jordan block.

$$N = J_2(0),$$

Take $B = I$.

4. Geometric multiplicity 1 is below algebraic multiplicity 2, preventing an eigenbasis. Equal eigenvalues do not imply a scalar multiple of the identity.

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CH08-E7

1. With the specified left cyclic coordinate shift P:

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

For each $v_\lambda = (1, \lambda, \lambda^2, \lambda^3)^T$,

$$Pv_\lambda = (\lambda, \lambda^2, \lambda^3, 1)^T = \lambda v_\lambda$$

because $\lambda^4 = 1$.

2. Suppose

$$\sum c_\lambda v_\lambda = 0,$$

The four coordinate equations form a Vandermonde system at distinct roots, with nonzero determinant. Equivalently, eigenvectors for distinct eigenvalues are independent. Thus all coefficients vanish.

3.

$$Cv_\lambda = (c_0 + c_1\lambda + c_2\lambda^2 + c_3\lambda^3)v_\lambda.$$

These polynomial evaluations at the four roots are the diagonal entries in the Fourier basis.

4. The complex eigenvalues occur in conjugate pairs. Here the spectrum is $1, i, -1, -i$. Pairing conjugate eigenvectors yields a real sine-cosine representation.

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CH08-E8

1.

$$B_\epsilon = \begin{bmatrix} 1 & 1 \\ 0 & \epsilon \end{bmatrix}, \qquad \det B_\epsilon = \epsilon.$$

The determinant is nonzero for $\epsilon > 0$, proving a basis.

2. Columns

$$b_1 = (1, 0)^T, \qquad b_2 = (1, \epsilon)^T.$$

$$B_\epsilon^T B_\epsilon = \begin{bmatrix} 1 & 1 \\ 1 & 1 + \epsilon^2 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_\pm = \frac{2 + \epsilon^2 \pm \sqrt{4 + \epsilon^4}}{2}.$$

Therefore

$$\kappa_2(B_\epsilon) = \sqrt{\frac{2 + \epsilon^2 + \sqrt{4 + \epsilon^4}}{2 + \epsilon^2 - \sqrt{4 + \epsilon^4}}}.$$

As $\epsilon \to 0^+$, $\lambda_- \to 0$ while $\lambda_+ \to 2$, so the condition number diverges. For numerical evaluation, use $\lambda_- = \epsilon^2 / \lambda_+$ to avoid subtractive cancellation.

3. Angle:

$$\cos\theta = \frac{b_1^T b_2}{\|b_1\| \|b_2\|} = \frac{1}{\sqrt{1 + \epsilon^2}}.$$

The angle tends to zero as $\epsilon \to 0^+$.

4. A coordinate perturbation δc gives

$$\|B_\epsilon \delta c\| \le \|B_\epsilon\|_2 \|\delta c\|.$$

If $\|\delta c\|_2 \le 10^{-8}$, reconstruction error is at most $\sqrt{\lambda_+}\, 10^{-8}$. If each coordinate error is bounded by 10^{-8}, the bound is $\sqrt{2\lambda_+}\, 10^{-8}$. In the reverse conversion, errors can be amplified by $\|B_\epsilon^{-1}\|_2 = 1/\sqrt{\lambda_-}$, which diverges as $\epsilon \to 0^+$. Exact invertibility does not guarantee numerically reliable coordinates.

CH08-E9

1.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix}, \qquad N = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}.$$

2. Since $(S \circ T)(v) = S(T(v))$, apply A first, then N.

$$NA = \begin{bmatrix} 3 & 1 \\ -3 & -1 \end{bmatrix}.$$

3. $T(1,2) = (3,-1,2)$, so $S(3,-1,2) = (5,-5)^T$.

$$(S \circ T)(1,2) = \begin{bmatrix} 3 & 1 \\ -3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \end{bmatrix}.$$

Both computations agree.

CH08-E10

Since $D(1) = 0$, $D(x) = 1$, and $D(x^2) = 2x$,

$$[D]_\beta = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus

$$[D]_\gamma = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Directly, $D(1+x) = 1$, and $D(1+x^2) = 2x = -2 + 2(1+x)$. For $[p]_\gamma = (1,2,1)^T$, $p = 4 + 2x + x^2$, so $D(p) = 2 + 2x = 2(1+x)$, whose γ-coordinates $(0,2,0)^T$ match matrix multiplication.

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References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 8.

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